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TAVR in Patients with Moderate Aortic Stenosis and Heart Failure

Brief/Cover title: TAVR in Moderate Aortic Stenosis and Heart Failure

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ABBREVIATIONS

AS = aortic stenosis

GDMT = guideline-directed medical therapy

LV = left ventricle

KCCQ-OS = Kansas City Cardiomyopathy Questionnaire Overall Summary score

TAVR = transcatheter aortic valve replacement

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In the evolving landscape of valvular heart disease treatment, moderate aortic stenosis (AS) in conjunction with heart failure and reduced ejection fraction (HFrEF) represents a critical clinical conundrum. This subgroup is often underdiagnosed and undertreated, but the *TAVR UNLOAD* trial,[1] published in this issue of *JACC,* and others like it are drawing attention to the need for earlier interventions, such as transcatheter aortic valve replacement (TAVR), as an alternative to watchful waiting and medical therapy. The editorial seeks to evaluate the clinical and research importance of this patient population, with emphasis on how TAVR may alter outcomes in patients with moderate AS and low ejection fraction.

The Landscape of Moderate AS and HFrEF

Current guidelines recommend aortic valve replacement for patients with severe AS presenting with symptoms, left ventricular (LV) ejection fraction <50%, or the presence of another Class 1 indication for open heart surgery. The treatment of moderate AS is mentioned only once in the most recent guideline recommendations—when another Class 1 indication for open heart surgery exists, such as the presence of significant coronary artery disease requiring coronary artery bypass grafts or the need for another valve replacement/repair.[2] Due to inherent limitations of echocardiography, deciphering between truly severe AS versus moderate AS could be challenging. Those challenges are amplified in the presence of reduced LV ejection fraction, when a low-flow state exists and when a lack of concordance between key echocardiogram measurements (peak velocity, aortic valve area, mean gradient) are present.

Recently, several retrospective studies have shown moderate AS to be associated with increased mortality, especially when reduced LV ejection fraction was present.[3-8] More importantly, some studies demonstrated potential benefits of early intervention among patients with moderate AS and reduced LV ejection fraction[4,8]; however, those studies suffered from

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multiple limitations germane to observational studies. To date, no randomized trials have been conducted among patients with moderate AS.

The *TAVR UNLOAD* Trial: Setting a Precedent

In the current issue of *JACC,* Van Mieghem et al[1] present the results of TAVR UNLOAD, the first randomized trial related to the treatment of moderate AS with reduced LV ejection fraction (20%-50%). This trial. Among 178 patients randomized to a strategy of early intervention with transfemoral TAVR with a balloon-expandable valve on top of optimal guideline-directed medical therapy (GDMT) vs optimal GDMT alone, the results were mixed after a median follow-up of 23 months: (i) TAVR UNLOAD failed to demonstrated the superiority of a strategy of early TAVR compared with GDMT alone in regard to the primary end point (hierarchical occurrence of all-cause death, disabling stroke, disease-related hospitalizations and heart failure equivalents, and change from baseline in the Kansas City Cardiomyopathy Questionnaire Overall Summary score [KCCQ-OS]): win ratio 1.31 (95%CI 0.91 to 1.88, p=0.14); (ii) TAVR plus GDMT showed greater improvement in the KCCQ-OS score compared with GDMT alone; and (iii) no increase in mortality or stroke was demonstrated with a strategy of early TAVR.

The investigators deserve commendation for completing the first randomized trial evaluating the treatment of moderate AS with aortic valve replacement, a previously uncharted territory. This ambitious study faced considerable challenges, particularly the COVID-19 pandemic, which significantly impacted recruitment. Initially targeting 600 patients, the trial was scaled down to 178 due to slow enrollment and funding constraints, which reduced its statistical power to detect meaningful differences. Moreover, the original design called for a primary analysis at one year; however, the follow-up period was extended, diluting the ability to capture

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early differences in clinical outcomes. This is important because there is discrepancy in the primary endpoint results when comparing the one-year follow-up with the longer, median 23-month follow-up. At one year, the trial showed a positive result, with TAVR outperforming GDMT in the ranked ordinal comparison (48% vs. 30.9%; win ratio 1.55 [95% CI, 1.04-2.31]), driven by improvements in quality of life. However, as follow-up extended to a median of 23 months, the initial positive findings were not sustained, and the trial ultimately failed to show superiority for the composite primary endpoint. A key factor was early crossover in 43% of patients in the control group at a median of 12 months (interquartile range 6-20), which likely diminished differences between the groups. This early intervention and the lower-than-expected event rates further reduced the trial's ability to demonstrate clear advantages of early TAVR. Moreover, while the trial demonstrated improvements in quality of life, these findings should be interpreted cautiously, as the trial lacked a sham-control. Without a placebo control, the reported quality-of-life gains may be influenced by patient expectations or psychological effects, making it harder to isolate the actual impact of TAVR on quality of life. A sham-controlled trial would help provide more definitive clarity on these outcomes. Further, it is important to note that the study was conducted exclusively with balloon-expandable valves, and it is unclear whether safety of TAVR in this population can be extrapolated to self-expanding valves, which may be associated with higher rates of pacemaker implantation that could be problematic in this high-risk population.[9]

The Road Ahead

Where does this leave us in managing moderate AS with reduced ejection fraction? The *TAVR UNLOAD* trial provides important insights, demonstrating the safety of early TAVR and its potential for improving quality of life. This improvement, paired with the lack of harm, may

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suffice to individualize the role of TAVR in the presence of moderate AS with reduced LV function, especially among patients with recurrent heart failure episodes and failure of optimal GDMT. One could argue that TAVR, in these circumstances, could represents the “fifth” pillar of heart failure therapy.

However, the broader question of whether preemptive TAVR in patients with moderate AS and preserved LV function improves long-term prognosis remains unanswered. Two larger ongoing trials (PROGRESS [NCT04889872] and EXPAND II [NCT05149755]) promise to shed light on this important issue. We owe a debt of gratitude to the TAVR UNLOAD investigators for their perseverance, as their efforts have opened the door to a new era of preventive treatment of AS.

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